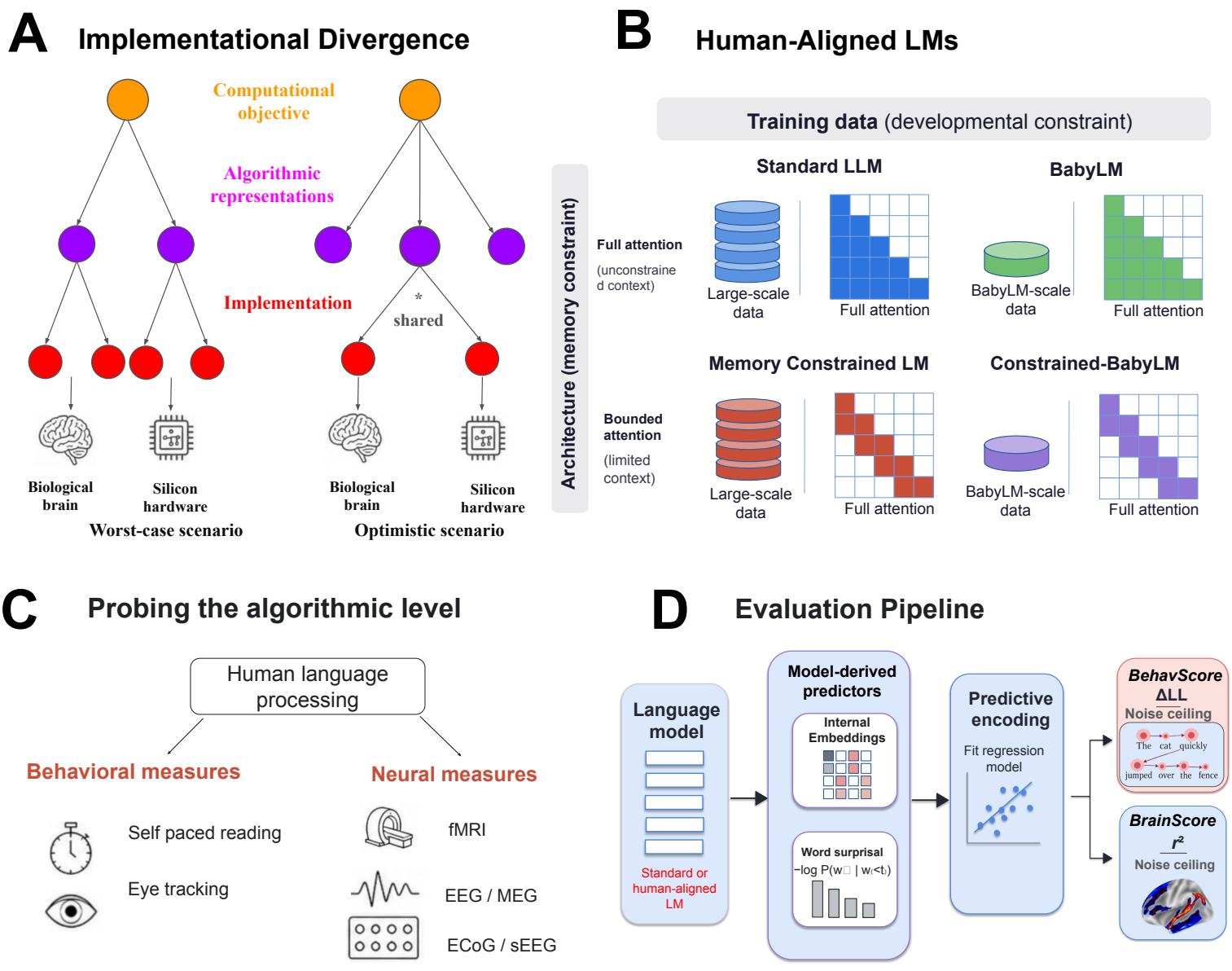

A
Implementational Divergence
Computational objective
Algorithmic representations
Implementation
shared
Biological brain
Silicon hardware
Worst-case scenario
Biological brain
Silicon hardware
Optimistic scenario
B
Human-Aligned LMs
Training data (developmental constraint)
Architecture (memory constraint)
Standard LLM
BabyLM
Full attention
(unconstrained context)
Large-scale data
Full attention
BabyLM-scale data
Full attention
Memory Constrained LM
Constrained-BabyLM
Bounded attention
(limited context)
Large-scale data
Full attention
BabyLM-scale data
Full attention
C
Probing the algorithmic level
Human language processing
Behavioral measures
Neural measures
Self paced reading
Eye tracking
fMRI
EEG / MEG
ECoG / sEEG
D
Evaluation Pipeline
Language model
Standard or human-aligned LM
Model-derived predictors
Internal Embeddings
Word surprisal
−log P(w□ | w<t)
Predictive encoding
Fit regression model
BehavScore
ΔLL
Noise ceiling
The cat quickly
jumped over the fence
BrainScore
r²
Noise ceiling

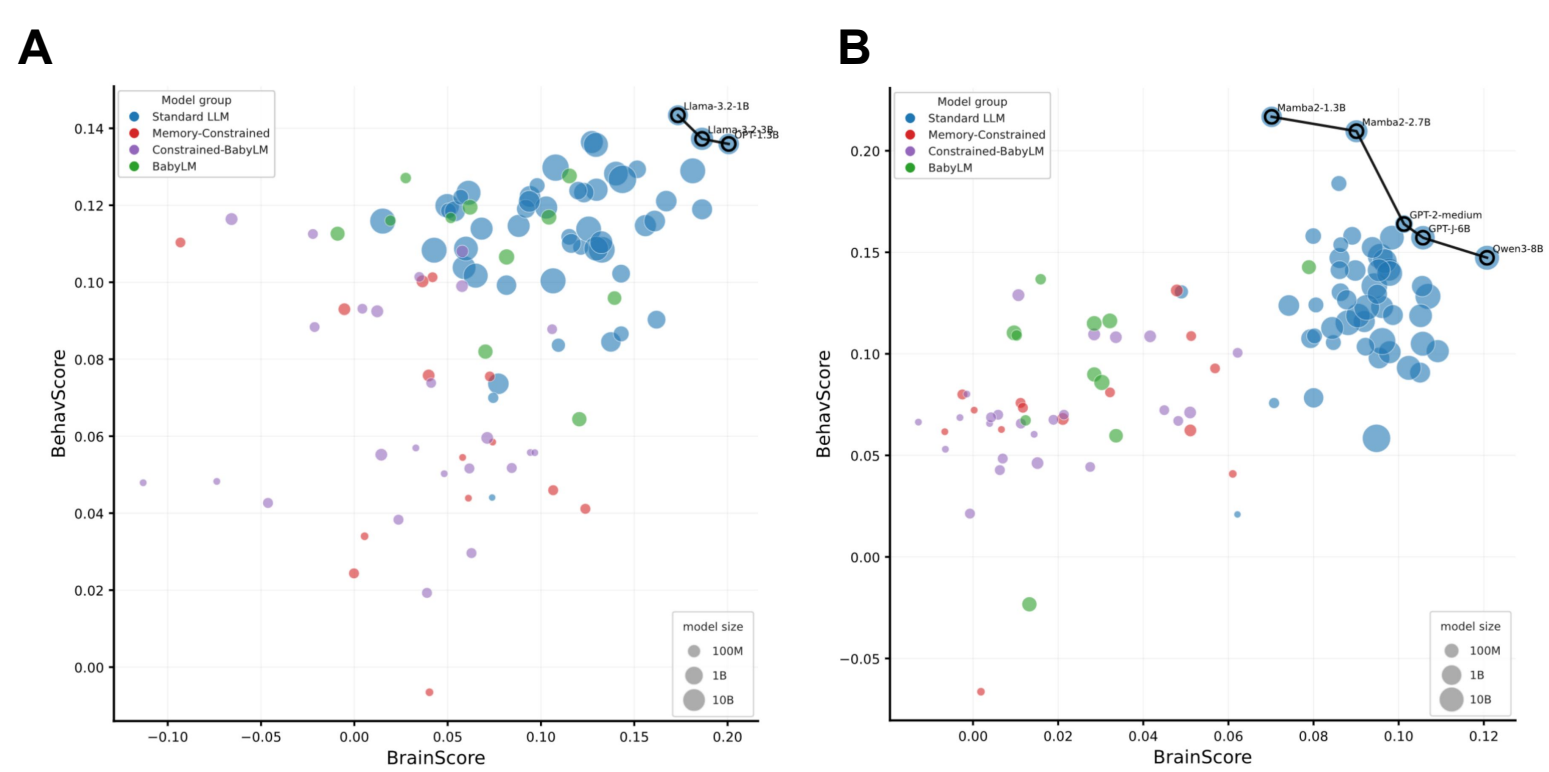
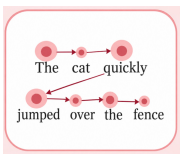


Fig 2: left: 0.09, right: 8.07, top: 0.13, bottom: 4.13



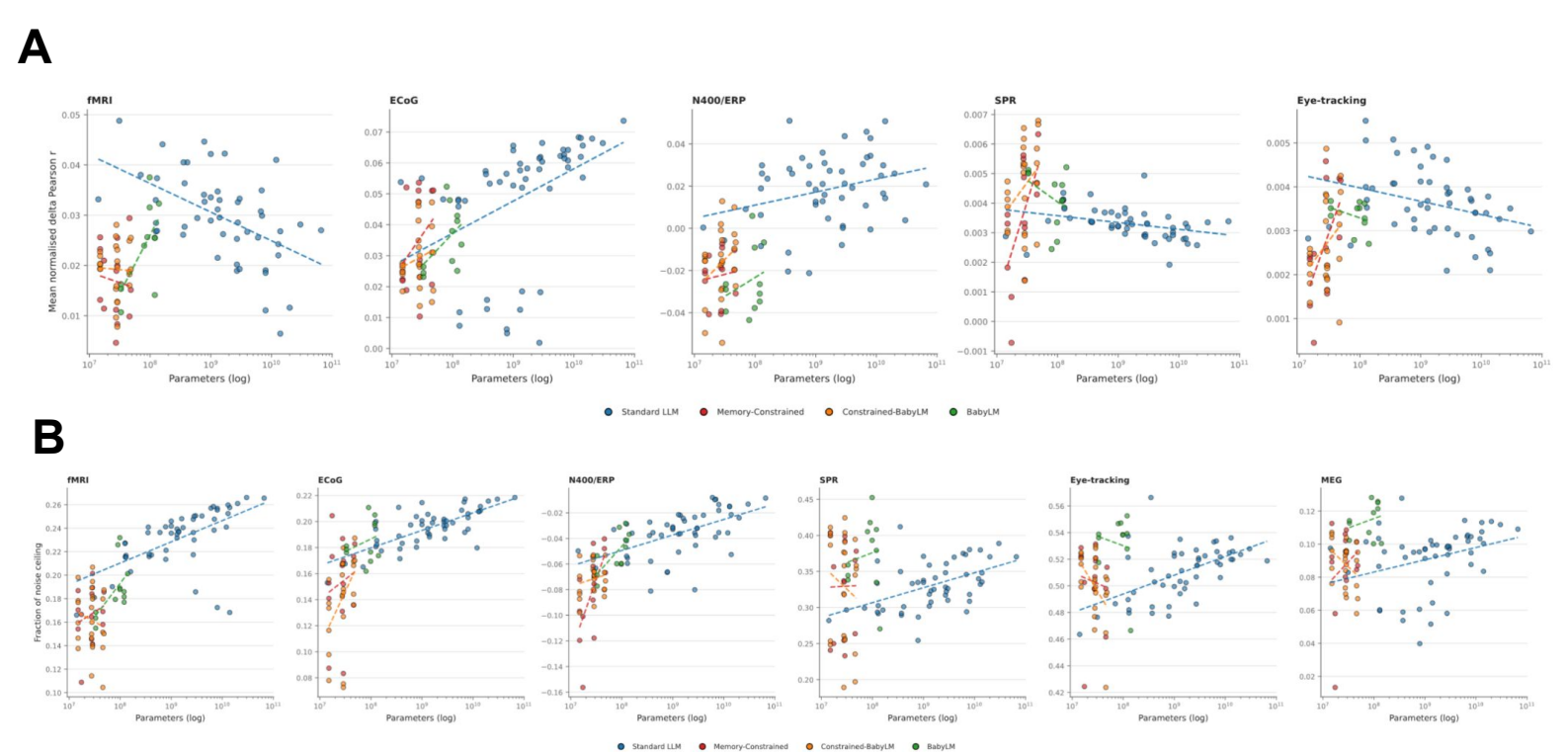


Fig 3: left: 0.07, right: 8.18, top: 0.13, bottom: 4.07

ROI and layer/head-wise analyses

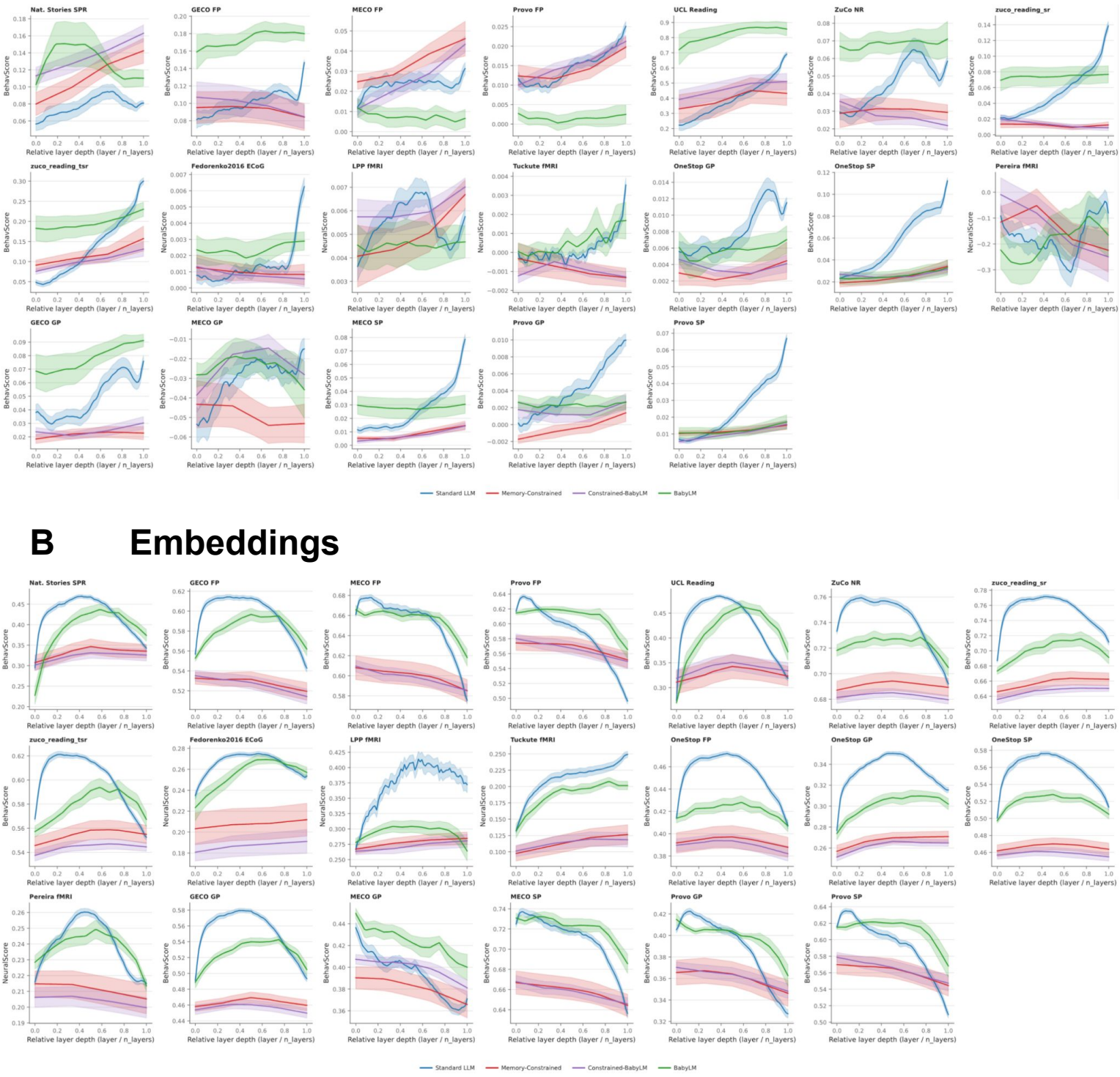
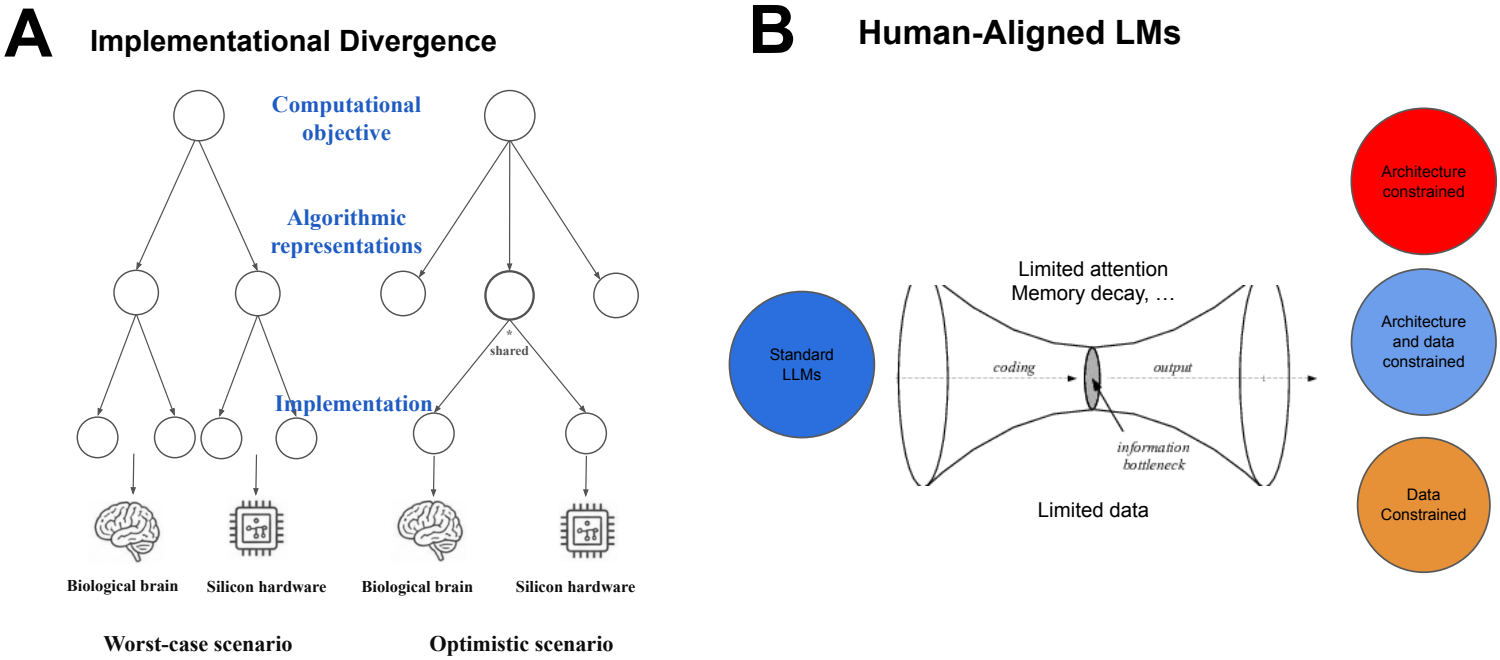


Fig 4: left: 0.09, right: 8.18, top: 0.58, bottom: 8.75

Appendix


A
Implementational Divergence
Computational objective
Algorithmic representations
Implementation
shared
Biological brain
Silicon hardware
Biological brain
Silicon hardware
Worst-case scenario
Optimistic scenario
B
Human-Aligned LMs
Standard LLMs
Limited attention
Memory decay, …
coding
output
information bottleneck
Limited data
Architecture constrained
Architecture and data constrained
Data Constrained
C
Probing the algorithmic level
Human language processing
Behavioral measures
Neural measures
Self paced reading
Eye tracking
fMRI
EEG / MEG
ECoG
D
Empirical Hypothesis Testing
LLM-derived predictors
Internal representations
Word surprisal
−log P(w□ | w<t)
Predictive Encoding models
Behavioral data
Self paced reading · Eye tracking
Brain data
fMRI · EEG · ECoG
Brain Score = R² (Y, Ŷ)
Predictive power
Behavior
Brain